



Contents lists available at ScienceDirect

Personality and Individual Differences

journal homepage: www.elsevier.com/locate/paid



Review

Network-based meta-analysis of sensory processing sensitivity: Exploring its relations with personality and temperament traits

Renjiao Tang^{a,b}, Xin Dai^a, Corina U. Greven^{c,d,e}, Zhi Li^f, Chunhong Zhu^a, Ni Yan^{a,*}

^a Faculty of Psychology, Southwest University, Chongqing, China

^b Department of Family Social Science, University of Minnesota, Saint Paul, United States

^c Department of Cognitive Neuroscience, Cognition and Behaviour, Donders Institute for Brain, Radboud University Medical Center, Nijmegen, the Netherlands

^d Karakter Child and Adolescent Psychiatry University Center, Nijmegen, the Netherlands

^e Psychology & Neuroscience, Institute of Psychiatry, Social, Genetic & Developmental Psychiatry Centre, King's College London, London, UK

^f Mt. Hope Family Center, University of Rochester, United States

ARTICLE INFO

Keywords:

sensory processing sensitivity
personality
temperament
environmental sensitivity
meta-analysis
network analysis

ABSTRACT

Sensory Processing Sensitivity (SPS) is a personality trait characterized by heightened sensitivity to both positive and negative environmental influences. This meta-analysis examines whether SPS constitutes a unique trait by analyzing its associations with key personality and temperament traits, while exploring age (children/adolescents vs. adults) and regional (Europe, Asia, North America) moderators. Analyzing 81 studies ($N = 59,543$), a network-based meta-analysis revealed SPS as an independent module. Its subdimensions showed distinct patterns: Ease of Excitation (EOE) and Low Sensory Threshold (LST) strongly correlated with neuroticism, negative affectivity, and the Behavioral Inhibition System (BIS), whereas Aesthetic Sensitivity (AES) correlated with openness, positive affectivity, and the Behavioral Activation System (BAS). Moderation analyses indicated these patterns varied by age and cultural region, with each subgroup displaying unique network configurations. These findings highlight SPS as a unique environmental sensitivity trait both “dark” and “bright” dimensions, urging further research into its developmental and cultural dynamics.

1. Introduction

The Environmental Sensitivity Framework posits that individuals differ in environmental perception and processing, with some being more sensitive than others (Greven et al., 2019; Pluess, 2015). Within this framework, sensory processing sensitivity (SPS) refers to a trait characterized by deeper cognitive processing of stimuli and increased emotional responsiveness (Aron & Aron, 1997). Individuals high in SPS exhibits dual reactivity—greater vulnerability in adversity but also enhanced benefits in supportive environments (Lionetti et al., 2019; Redfearn et al., 2020; Slagt et al., 2018). However, the extent to which SPS represents a distinct environmental sensitivity trait remains debated, due to its overlap with other sensitivity traits like personality and temperament (Bröhl et al., 2022; Greven et al., 2019; Lionetti et al., 2019). Clarifying this distinction is crucial for advancing research on environmental sensitivity and establishing SPS as a unique construct beyond existing personality or temperament traits.

This study employs both effect size and network-based meta-analysis

to examine how SPS is associated with other personality and temperamental traits while maintaining its distinct identity in the sensitivity network. By mapping the relations between SPS and the most comprehensive range of personality and temperament traits to date, the study aims to study overlap and differences between SPS and established personality and temperament constructs. This approach helps clarify SPS's unique contributions in shaping individuals' responses to environmental stimuli, going beyond what is captured by personality and temperament alone. Additionally, while much of previous research has focused on pairwise correlations, this study considers factors such as age and culture as potential moderators of these relations, which may explain the variability across findings (Lionetti et al., 2019; Weyn et al., 2021). Collectively, these efforts aim to advance the understanding of SPS within Environmental Sensitivity Framework and shed light on the complex interplay between individuals and environmental contexts.

* Corresponding author at: Faculty of Psychology, Southwest University, 2 Tiansheng Str., Beibei, Chongqing 400715, China.

E-mail address: niyan@swu.edu.cn (N. Yan).

<https://doi.org/10.1016/j.paid.2025.113435>

Received 14 May 2025; Received in revised form 23 August 2025; Accepted 26 August 2025

Available online 7 September 2025

0191-8869/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1.1. SPS as a marker for environmental sensitivity

SPS was initially conceptualized as a unidimensional trait capturing individuals' sensitivity to environmental stimuli, and it was validly measured using the 27-item Highly Sensitive Person Scale (Aron & Aron, 1997). Subsequent factor-analytic work has identified a three-factor model within the Highly Sensitive Person Scale (HSPS) item pool: (1) ease of excitation (EOE), referring to being easily overwhelmed by internal and external stimuli; (2) low sensory threshold (LST), indicating unpleasant sensory arousal to external stimuli; and (3) aesthetic sensitivity (AES), reflecting pleasure from aesthetic experiences and positive stimuli (Pluess et al., 2018; Smolewska et al., 2006). It is crucial to acknowledge, however, that this three-factor model has been established almost exclusively through the HSPS. This framework has since proliferated across the field and has been translated and adapted for use in numerous cultural contexts, generating over 30 variants for adult, child self-report, and parent-report formats across more than 12 countries—from Europe to Asia (e.g. De Gucht et al., 2023; Şengül-İnal & Sümer, 2020; Yano et al., 2020). While this three-factor structure is now widely used to guide research, the extent to which this model generalizes beyond the HSPS to represent the core structure of the SPS trait requires validation with alternative measurement tools. Therefore, the present meta-analysis synthesizes the substantial empirical literature built upon this prevailing operationalization. Our aim is to provide the most comprehensive map to date of how the SPS trait, as currently understood and measured, relates to other personality and temperament constructs. This approach enables us to situate SPS within the broader dispositional landscape while identifying key priorities for future construct validation.

Empirical evidence supports individuals with high SPS are more sensitive to environmental influences, experiencing greater maladjustment in adverse environments (Andresen et al., 2018). Conversely, these individuals have been found to derive greater benefits from positive environmental stimuli, as demonstrated in studies utilizing positive mood induction tasks and intervention programs (Nocentini et al., 2018; Pluess & Belsky, 2013; Pluess & Boniwell, 2015). Similarly, SPS has been shown to correlate with other personality and temperament traits associated with both negative and positive experiences. Research indicates that SPS is positively associated with neuroticism, negative affectivity, openness, and positive affectivity, again reflecting both positive and negative susceptibility (Ahadi & Basharpour, 2010; Aron et al., 2005; Smolewska et al., 2006). These correlations suggest that the differential susceptibility observed in high-SPS individuals may be partially explained by their alignment with other personality and temperament traits, such as neuroticism and openness. This highlights the need for further nuanced research into the complex interplay between SPS and other environmental sensitivity traits.

1.2. SPS and other traits linked to environmental sensitivity

SPS is a distinct environmental sensitivity trait with its unique structure, yet it overlaps with personality and temperament traits and cannot be fully explained by them (Lionetti et al., 2018; Pluess et al., 2018). While prior research has explored some of these associations, this study is the first to comprehensively examine both the overlaps and distinctions between SPS and a broad range of personality and temperament traits, addressing a key gap in understanding environmental sensitivity.

1.2.1. Personality traits

Personality refers to the characteristic pattern of thinking, feeling, and behaviors that remain relatively consistent over time and across contexts (Allport, 1961). Excluding the dimensions of agreeableness and conscientiousness—traits that show no consistent relations with SPS (Lionetti et al., 2019; Pluess et al., 2018; Smolewska et al., 2006)—we focus on neuroticism, extroversion, and openness, all of which

demonstrate more consistent associations with SPS. First, neuroticism, which reflects emotional instability, shows a robust positive correlation with SPS ($r = 0.31\text{--}0.50$, Weyn et al., 2021; Lionetti et al., 2019; Smolewska et al., 2006; Pluess et al., 2018), highlighting its role in understanding SPS's negative reactivity. Second, extroversion is negatively associated with SPS total score and the subdimensions of EOE and LST, aligning with high SPS individuals' introverted tendencies (Lionetti et al., 2019; Pluess et al., 2018; Weyn et al., 2021). Third, openness shows a weak overall association with SPS ($r = 0.13\text{--}0.19$) but a stronger positive correlation with AES ($r = 0.23\text{--}0.47$) (Lionetti et al., 2019; Pluess et al., 2018; Weyn et al., 2021). Despite these established relations, the significant heterogeneity persists across studies, with correlation coefficients ranging widely and producing inconsistent conclusions (Şengül-İnal et al., 2018; Yano et al., 2020; Weyn et al., 2021). Therefore, further exploration is needed to examine these associations, especially the differences between the subdimensions of SPS and personality constructs, as this could enhance our understanding of SPS in the context of existing knowledge about personality.

1.2.2. Temperament traits

As a multidimensional construct, temperament encompasses traits such as positive affectivity, negative affectivity, and effortful control (Rothbart's theory; Rothbart & Jones, 1998), along with the Behavioral Inhibition System (BIS) and Behavioral Activation System (BAS) from Gray's reinforcement sensitivity theory (Gray, 1982). These temperamental traits capture physiological and emotional sensitivities to environmental influences and have been empirically correlated with SPS.

Temperamental traits related to emotional tendencies, such as negative affectivity and positive affectivity, are associated with SPS. First, SPS total score, as well as its subdimensions of EOE and LST, are positively correlated with negative affectivity ($r = 0.21\text{--}0.5$) (Pluess et al., 2018; Weyn et al., 2021), whereas AES shows negligible associations ($r = -0.1\text{--}0.05$) (Lionetti et al., 2019; Sobocko & Zelenski, 2015). Positive affectivity shows greater variability: AES consistently demonstrates positive correlations ($r = 0.16\text{--}0.29$, Weyn et al., 2021; Pluess et al., 2018), while EOE and LST display mixed results, ranging from negative ($r = -0.56 \sim -0.11$; Weyn et al., 2021) to positive ($r = 0.14\text{--}0.41$; Pluess et al., 2018). Second, temperamental traits like effortful control, BIS, and BAS capture individuals' ability to regulate and adjust their responses to environmental stimuli. Specifically, BIS shows robust positive associations with SPS and its subdimensions (Pluess et al., 2018; Smolewska et al., 2006; Sobocko & Zelenski, 2015), supporting its conceptual alignment with SPS's "pause-to-check" mechanism (Aron & Aron, 1997). In contrast, SPS's association with effortful control varies widely ($r = -0.40\text{--}0.48$, Weyn et al., 2021; Pluess et al., 2018), as does association with BAS ($r = -0.15\text{--}0.42$, Sobocko & Zelenski, 2015; Pluess et al., 2018; Şengül-İnal et al., 2018). The substantial heterogeneity across studies underscores the need for a meta-analysis to clarify how specific temperament dimensions relate to different facets of SPS.

1.3. Moderators between SPS and personality/temperament traits

The relation between SPS and other personality and temperament traits may shift as individuals progress through different developmental stages and vary across different cultural regions contexts. This study identifies two key moderators in these relations: age and region.

1.3.1. Age

The role of age in moderating the relations between SPS and other personality/temperament traits remains debated. SPS is generally considered stable across development with genetic underpinnings (Greven et al., 2019), with consistent associations with traits like neuroticism and negative affectivity in both children and adults (Lionetti et al., 2019). However, emerging evidence suggests age-related variations, particularly when examining specific SPS subdimensions. For

instance, as individuals transition from childhood to adulthood, their cognitive processes become more refined, enhancing the ability to process complex stimuli, which may lead to heightened emotionality and openness to environmental stimuli, particularly among high SPS individuals (Acevedo, 2020). This is reflected in findings where a moderate significant positive correlation between AES and neuroticism is observed in adults ($r = 0.38$; Gerstenberg, 2012), but this relation does not hold in children and adolescents ($r = 0.07$, $M_{age} = 10.77$; Weyn et al., 2021; $r = -0.00$, $M_{age} = 17.06$; Pluess et al., 2018). Furthermore, the AES-openness correlation is also notably stronger in adults than children (Lionetti et al., 2019). Taken together, conducting a meta-analysis to investigate whether SPS is moderated by age would be essential in clarifying these inconsistencies and providing a more comprehensive understanding of the potential developmental modulation of SPS.

1.3.2. Region

Although regional differences in the relations between SPS and other traits remain underexplored, emerging evidence suggests both genetic and sociocultural factors may contribute to such variation. First, the 5-HTTLPR short allele, associated with heightened susceptibility to negative environments and life events, is more prevalent in Eastern populations (e.g., 80 % in Japanese vs. 40 % in Americans; Chiao & Blizinsky, 2010), potentially strengthening SPS's associations with traits reflecting negative sensitivity in Eastern samples (Belsky & Pluess, 2009; Homberg et al., 2016; Pluess, 2015). Second, cultural norms may further shape the expression of sensitivity. While Western cultures emphasize individual expression, collectivist cultures, such as those in East Asia, prioritize group harmony and interdependence, often encouraging the suppression of individual emotions for the sake of interpersonal meaning (Matsumoto et al., 2008). Therefore, highly sensitive individuals in Western contexts may express their emotional sensitivity more openly, whereas their Eastern counterparts may suppress emotions that differ from the group. These contrasting cultural norms suggest potential differences in how sensitivity is expressed across Eastern and Western populations. Cross-cultural studies support these hypotheses. For instance, Japanese adolescents showed a negative relation between SPS and positive affectivity and showed distinct patterns in SPS-BAS associations, contrary to Western findings (Pluess et al., 2018). These discrepancies highlight the influence of both genetic and cultural factors in shaping the expression of sensitivity traits. Despite these findings, to our knowledge, no meta-analysis has yet examined the regional differences in the relation between SPS and personality/temperament traits, highlighting the need for further investigation in this area.

1.4. The current study

This study aims to clarify how SPS fits within the broader landscape of established personality and temperament traits through a meta-analysis of the empirical literature. To do so, we synthesize the available quantitative literature, which almost exclusively uses the HSPS and its derivatives to operationalize the construct. The theoretical value of our analysis lies not in asserting SPS' uniqueness per se, but rather in examining whether the specific pattern of associations predicted by SPS theory holds. First, the effect size meta-analysis will synthesize existing research to create a correlation matrix for SPS (including its subdimensions of EOE, LST, AES), and key personality and temperament traits, while exploring age and region as potential moderators. The moderation analyses remain exploratory in nature due to limited evidence. Second, the network analysis will leverage the synthesized correlation matrices to visualize and analyze the structural relations among SPS, personality, and temperament traits. This method will allow us to test whether SPS forms an independent trait cluster, distinct from traditional personality and temperament dimensions. We hypothesize that SPS will demonstrate related but distinct patterns from personality and temperament traits, forming an independent network module.

Specifically, we expect its subdimensions—EOE, LST, and AES—to correlate differently with traits associated with negative and positive reactivity. Ultimately, this study aims to clarify the relations between SPS and other personality or temperament traits, with a specific focus on examining how sensitivity characteristics vary across different subgroups based on age (children/adolescents vs. adults) and region (Europe, North America, and Asia). These factors will be explored as key moderators to better understand how SPS manifests differently across developmental stages and cultural contexts.

2. Method

This research was preregistered (September 25, 2024, with subsequent amendments) at OSF. The study materials, data and analysis scripts used for this article can be accessed at https://osf.io/kda6m/?view_only=e0836a2aea814c12a7759beb8b3104fb.

2.1. Meta-analytic procedure

We conducted a systematic literature search across the PsycINFO, Web of Science, and PubMed databases, completed on May 25, 2023, using keyword combinations for SPS, personality, and temperament (see Appendix A for full search strings). After removing duplicates, 2921 articles remained. Articles were then screened based on titles and abstracts, followed by a full-text assessment for eligibility against predefined inclusion criteria. These criteria required studies to be quantitative, use psychometrically sound measures (Cronbach's $\alpha \geq 0.7$), report necessary demographic and correlational data, and focus on non-clinical populations. The final set comprised 81 eligible studies. The complete literature screening procedure is detailed in Appendix A and summarized in Fig. 1.

Data, including study characteristics and pairwise Pearson correlation coefficients for 12 variables of interest, were extracted by three trained postgraduates with a high degree of consistency (93 % agreement). A detailed description of the data extraction protocol is available in Appendix A. For analysis, following the Lancet Commission on adolescent health and well-being (Patton et al., 2016), age was dichotomized into children/adolescents (<20 years) and adults (≥ 20 years). Additionally, while European and North American populations are often collectively referred to as "Western" in previous studies, we further refined this classification. The 20 countries involved in this study were grouped into three geographical categories: Europe, Asia, and North America.

2.2. Analytical strategy

The calculation of effect sizes and the testing of moderating effects were conducted using Comprehensive Meta-Analysis 3 (CMA3), and network analysis was conducted in R statistical software (Version 4.3.1). Specifically, we first used CMA3 to calculate the overall effect sizes of the correlation coefficients between traits and then tested the moderating effects of age and region. Next, we constructed a comprehensive correlation effect sizes matrix and imported it into R for network community detection and visualization.

2.2.1. Effect size calculation

The meta-analysis method for correlation coefficient was adopted in this study, with the zero-order correlation coefficient (r) serving as the effect size metric. To obtain the pooled mean effect size for each pairwise correlation among the 12 traits of interest, we followed a two-step process.

First, the zero-order correlation (r_i) extracted from the studies were converted to Fisher's Z values. The weighted average of these Fisher Z values was calculated by assigning weights based on the inverse variance of each Z value, and then it was transformed back into a correlation coefficient to obtain the overall effect value (ρ). Second, we estimated

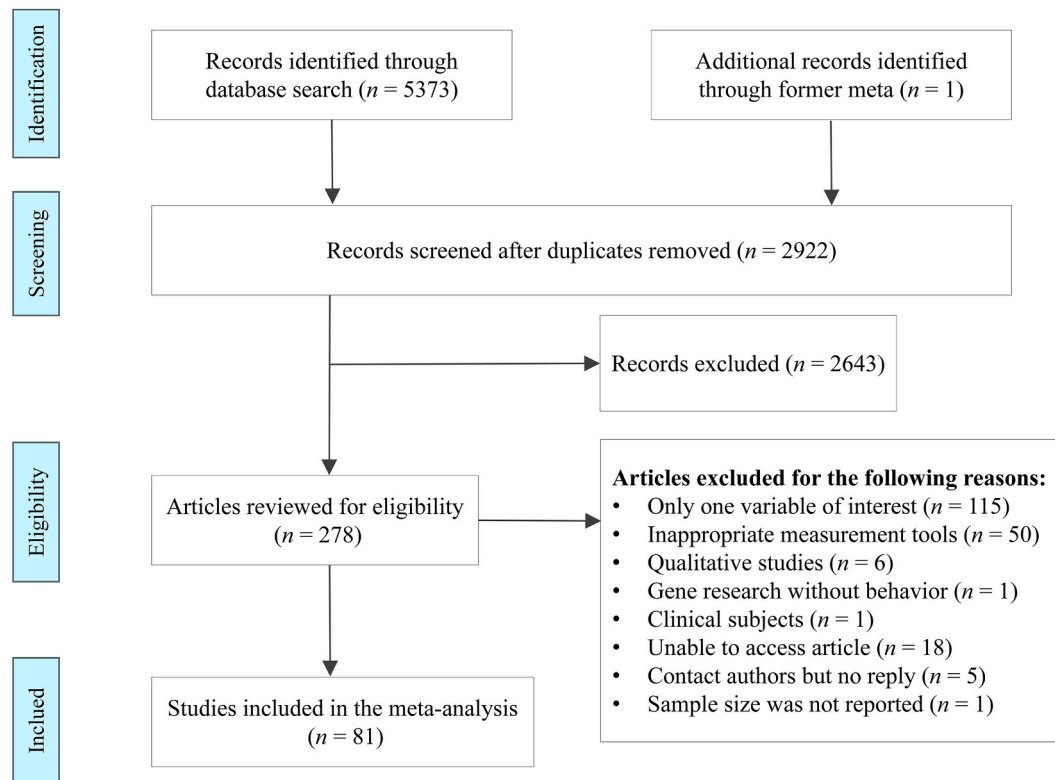


Fig. 1. Flow diagram (PRISMA) of the literature screening and selection process.

the 95 % confidence interval for the overall effect size using SEp as the standard deviation (Schmidt & Hunter, 2014). Detailed formulas are available in the Appendix A.

2.2.2. Heterogeneity analysis

When calculating effect sizes, it is essential to consider two alternative models: the fixed effects model and the random effects model. These models differ in how they compute effect size and its confidence interval. The purpose of the heterogeneity test was to assess the degree of dispersion or variability in effect sizes across studies. If heterogeneity was detected, the random effects model was applied; otherwise, the fixed effects model was used.

The Q test was used to assess whether the observed variability in effect sizes is greater than what would be expected by chance alone, thus indicating the presence of heterogeneity. Further, the I^2 index represents the degree of heterogeneity by quantifying the percentage of total variation across studies due to heterogeneity rather than random sampling error. A significant Q test ($p < 0.05$) indicates the presence of heterogeneity, and an I^2 value above 75 % suggests substantial variation between studies, indicating that systematic or heterogeneous variation rather than random sampling error explains most of the differences (Borenstein et al., 2009; Hedges & Vevea, 1998). Therefore, if the I^2 value was greater than 75 %, the random effect model was selected; otherwise, the fixed effect model was selected (Higgins et al., 2003).

2.2.3. Moderation analysis

Moderation analyses were conducted using CMA3 software. In CMA3, moderation analysis is performed by coding the potential moderator variables (e.g., age and region) as categorical variables. Specifically, we entered the effect sizes and corresponding variances into CMA3 and categorized the studies based on age groups (children/adolescents vs. adults) and geographical regions (e.g., Europe, Asia, North America). The software then calculated the pooled effect sizes for each subgroup and compared them to determine whether the moderator

variables influenced the overall effect size. The significance of the moderation was determined using the p -value associated with the Q_b -statistic (i.e., the between-group heterogeneity statistic), with a significant p -value indicating the presence of a moderating effect (Borenstein et al., 2009). It should be noted that the moderation analyses focused exclusively on SPS and its associations with personality and temperament, without conducting moderation analyses on the subdimensions of personality or temperament.

2.2.4. Constructing network matrix

To avoid potential multicollinearity within the network, the SPS total score, derived from the EOE, AES, and LST subdimensions, was excluded, leaving eleven sensitivity traits as nodes for the network analysis, resulting in an 11 by 11 positive definite correlation matrix for subsequent analysis. Additionally, in the adult, Asian, and North American population networks, no data was available on the correlations among the three temperament traits (positive affectivity, negative affectivity, effort control), their correlations with BIS/BAS, or the correlations between positive affectivity/effort control and SPS. Consequently, a complete correlation matrix could not be generated. To ensure a fully connected and comparable network for community detection, we excluded positive affectivity, negative affectivity, and effortful control. This exclusion led to the formation of five 8 by 8 moderated correlation matrices (for children/adolescents and adults, for Europe, Asia, and North America) for the subsequent community detection analyses.

2.2.5. Network community detection

In this network analysis, sensitivity traits were represented as nodes, and the correlation coefficients between them as edges. It is important to note that our approach differs from methods like Unique Variable Analysis (UVA), which identify redundant variables for potential removal prior to analysis (Christensen et al., 2020). As our input is a meta-analytic correlation matrix rather than raw data, our goal was not

to reduce variables but to map the structural position of the established SPS construct within a network of other well-defined traits. We used the EBICglasso method from the qgraph package to estimate the partial correlation matrix among all traits, applying a penalty to sparsify the network and exclude spurious connections (Epskamp et al., 2018). The penalty strength, controlled by the tuning parameter, was set to the default value of 0.5 (Epskamp et al., 2018). Community detection within the trait network was performed using the EGANet package (version 2.0.6; Golino & Christensen, 2024), which groups traits with strong positive partial correlations into modules. To formally test the stability of this community structure and determine the optimal number of modules, we employed a bootstrap procedure (500 replicates) within EGANet. This resampling method operates by repeatedly generating new samples from the original data and applying the community detection algorithm to each one. The consistency of the modules across these replicates is then used to generate a confidence interval for the optimal number of communities, providing a quantitative assessment of the solution's robustness (Christensen et al., 2020). To ensure the reliability of our findings, we conducted two additional robustness checks. First, we performed a sensitivity analysis on the EBICglasso tuning parameter. Following Epskamp and Fried (2018), we systematically re-estimated the networks using a range of alternative gamma values: $\gamma = 0$ (equivalent to BIC), $\gamma = 1.0$, and $\gamma = 2.0$, representing less and more conservative penalties for model complexity, respectively. Second, to further ensure the robustness of the identified modules, we also used the sna package (version 2.7–2; Butts, 2008) to apply the triangulated maximally filtered graph (TMFG) model and the Louvain community detection method, which provided convergent evidence for our primary findings.

3. Results

3.1. Sample description

This meta-analysis included data from 81 studies, encompassing 107 samples ranging in age from 2 to 59 years, yielding a total of 1020 effect

Table 1
Characteristics and source of effect sizes in included studies.

Studies information	Coding strategy	Descriptive results
<i>Characteristics of included studies</i>		
Publication type	Was coded as journal article ($n = 75$) articles or dissertations ($n = 6$)	
Published year	Was coded as the year of publication	Ranging from 1991 to 2023 ($M = 2014$)
Sample size	Was coded as the number of study participants	Sample group = 107; Sample size = 59,543
Sample distribution	Was coded by the continent of the study participants' countries	Europe = 36,180; North America = 12,692; Asia = 10,671
Age (year)	Was coded as child/adolescent or adult based on a 20-year threshold.	Ranging from 2.00 to 59 ($M = 21.88$, $SD = 10.88$)
Gender (% female)	Was coded as the percentage of female participants	Ranging from 42 to 93 ($M = 59.84$, $SD = 12.32$)
<i>Source of Effect Sizes in Included Studies</i>		
Effect size number within models	Was coded as SPS ($k = 198$), Personality ($k = 90$), Tem. ($k = 28$) and BIS/BAS ($k = 13$)	
SPS vs other models	Was coded as SPS vs Personality ($k = 281$); Tem. ($k = 69$); BIS/BAS ($k = 58$)	
Personality vs other models	Was coded as Personality vs Tem. ($k = 171$); BIS/BAS ($k = 100$)	
Tem. vs BIS/BAS	Was coded as the effect sizes of Tem. vs BIS/BAS ($k = 12$)	

Note. k , number of effect sizes; SPS, the sensory processing sensitivity model; Tem., the Rothbart's model of temperament.

sizes (see Table 1 and Table S1).

3.2. Overall effect sizes

The results of the heterogeneity tests, including the total sample size and the number of studies included for each effect size calculation, are presented in Table S2 (Appendix B). According to the criteria for heterogeneity classification (Higgins et al., 2003) and the interpretation of the I^2 index by Borenstein et al. (2009), we used a random effects model to calculate the pooled effect sizes of 60 raw pairwise correlation coefficients among the 12 traits. However, for four specific pairs (negative affectivity and openness, negative affectivity and BIS, positive affectivity and BIS, positive affectivity and LST), a fixed effect model was applied because their I^2 values were less than 70 %. In total, we obtained 64 overall mean weighted correlation effect sizes, where were used to form a positive definite matrix for subsequent network analysis.

3.2.1. Relations among SPS and its subdimensions

As shown in Fig. 2, the SPS total score was strongly positively correlated with EOE and LST, while showing a relatively lower correlation with AES. Similarly, EOE and LST were more closely related to each other, whereas both exhibited weaker correlation with AES.

3.2.2. Relations among SPS (and its subdimensions) and personality traits

As shown in Fig. 3, the SPS total score was positively correlated with neuroticism and openness, but negatively correlated with extraversion. At the subdimension level, distinct patterns emerged. Specifically, EOE and LST demonstrated a strong positive correlation with neuroticism, a small negative correlation with extraversion, and no correlation with openness. In contrast, AES was positively correlated with all three traits, showing weaker correlation with neuroticism and extroversion, but a stronger correlation with openness.

3.2.3. Relations among SPS (and its subdimensions) and temperament traits

As shown in Fig. 4, the SPS total score and its subdimensions were positively correlated with negative affectivity but showed no significant correlation with effortful control. Positive affectivity was not significantly correlated with the SPS total score, EOE, or LST, but showed a small correlation with AES. In terms of BIS/BAS, SPS and its subdimensions were positively correlated with both BIS and BAS. Specifically, BIS had a strong positive correlation with the SPS total score and EOE, and a moderate correlation with LST and AES. In comparison, BAS had the strongest correlation with AES, followed by the SPS total score and EOE, but showed no significant correlation with LST.

3.3. Moderating effect of age and region

After calculating and constructing the matrix of pooled effect sizes, we performed moderating analyses to investigate the discrepancies caused by age and region. The significant results are summarized in Table 2 (see Table S4 in Appendix B for complete results, including non-significant ones).

However, due to insufficient data, correlation estimates between several traits were unavailable under certain moderating conditions. Therefore, when specific correlations are not reported in the following results, it indicates a lack of data for those relations.

3.3.1. Relations among SPS and its subdimensions

The correlations between the SPS total score and its three subdimensions were not moderated by region and age. The only age difference was detected in the relation between EOE and LST: the correlation was significantly higher among adults ($r_{adult} = 0.63$) than children/adolescents ($r_{child/adolescent} = 0.45$), with a Q_b value of 18.07 ($p < 0.001$).

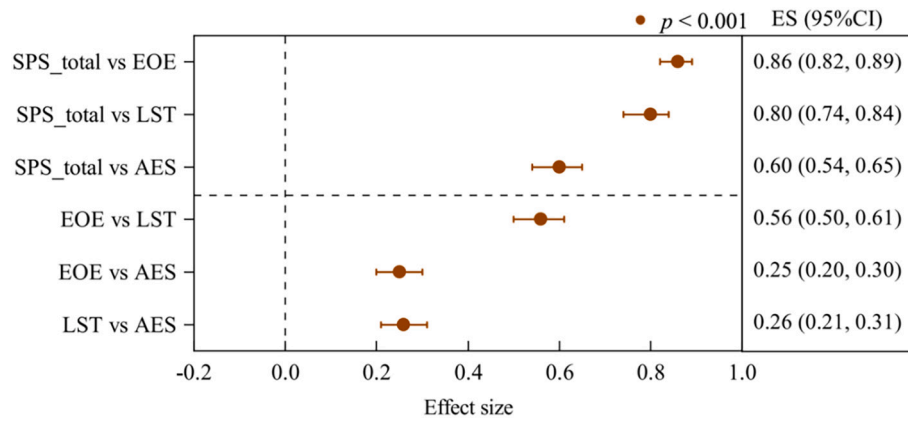


Fig. 2. The forest plot of the SPS inter-model.

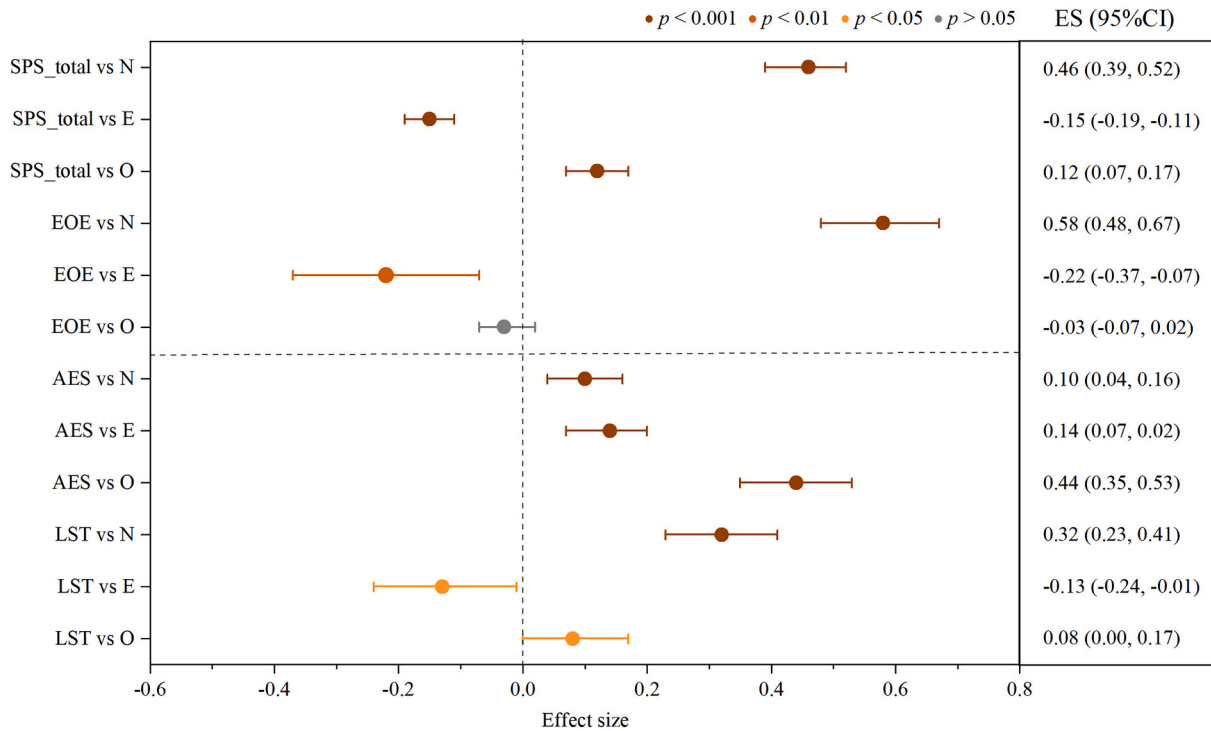


Fig. 3. The forest plot illustrating the correlation between the SPS model and personality traits.

3.3.2. SPS (and its subdimensions) and personality traits

Despite regional and age differences, the patterns of relations between SPS and personality traits were generally consistent. Neuroticism had stronger correlations with both the SPS total score ($r_{adult} = 0.52$ vs. $r_{child/adolescent} = 0.35$, $Q_b = 11.37$, $p < 0.001$) and EOE ($r_{adult} = 0.65$ vs. $r_{child/adolescent} = 0.37$, $Q_b = 13.57$, $p < 0.001$) in adults compared to children/adolescents. Openness also showed stronger correlations with the SPS total score ($r_{adult} = 0.17$ vs. $r_{child/adolescent} = 0.05$, $Q_b = 5.12$, $p < 0.05$) and AES ($r_{adult} = 0.54$ vs. $r_{child/adolescent} = 0.22$, $Q_b = 27.66$, $p < 0.001$) in adults.

Regionally, extraversion showed significant correlations with EOE in the European ($r = -0.29$) and North American ($r = -0.09$) groups, but not in the Asian group ($r = -0.03$), as indicated by a Q_b value of 16.11 ($p < 0.01$). Conversely, a significant positive correlation between extraversion and AES was observed in the Asian population ($r = 0.20$), which was stronger than in the European ($r = 0.13$), with no significant correlation found in the North American group ($r = 0.00$, $Q_b = 44.22$, $p < 0.001$).

3.3.3. SPS (and its subdimensions) and temperament traits

Age differences in the relations between SPS and temperament traits were most evident for BAS. The correlation between BAS and the SPS total score decreased with age, from $r_{child/adolescent} = 0.28$ to $r_{adult} = 0.11$ ($Q_b = 4.10$, $p < 0.05$). However, the correlation between BAS and LST increased with age, being non-significant in children ($r = -0.02$) but positive in adults ($r = 0.12$, $Q_b = 4.23$, $p < 0.05$).

Regionally, the strongest correlation between the SPS total score and negative affectivity was found in European groups ($r = 0.50$), followed by North American ($r = 0.44$), and Asian groups ($r = 0.30$) ($Q_b = 18.62$, $p < 0.001$). A similar trend was observed for EOE, where the correlation was stronger in Europeans ($r = 0.49$), Asians ($r = 0.31$) ($Q_b = 4.08$, $p < 0.05$). For AES, a significant correlation with negative affectivity was observed in Europeans ($r = 0.23$), while no significant relation was found in Asians ($r = 0.06$) ($Q_b = 7.36$, $p < 0.01$). LST correlated most strongly with negative affectivity in Europeans ($r = 0.36$), with a weaker correlation in Asians ($r = 0.09$, $Q_b = 5.88$, $p < 0.05$).

Additionally, EOE correlated most strongly with BAS in Europeans (r

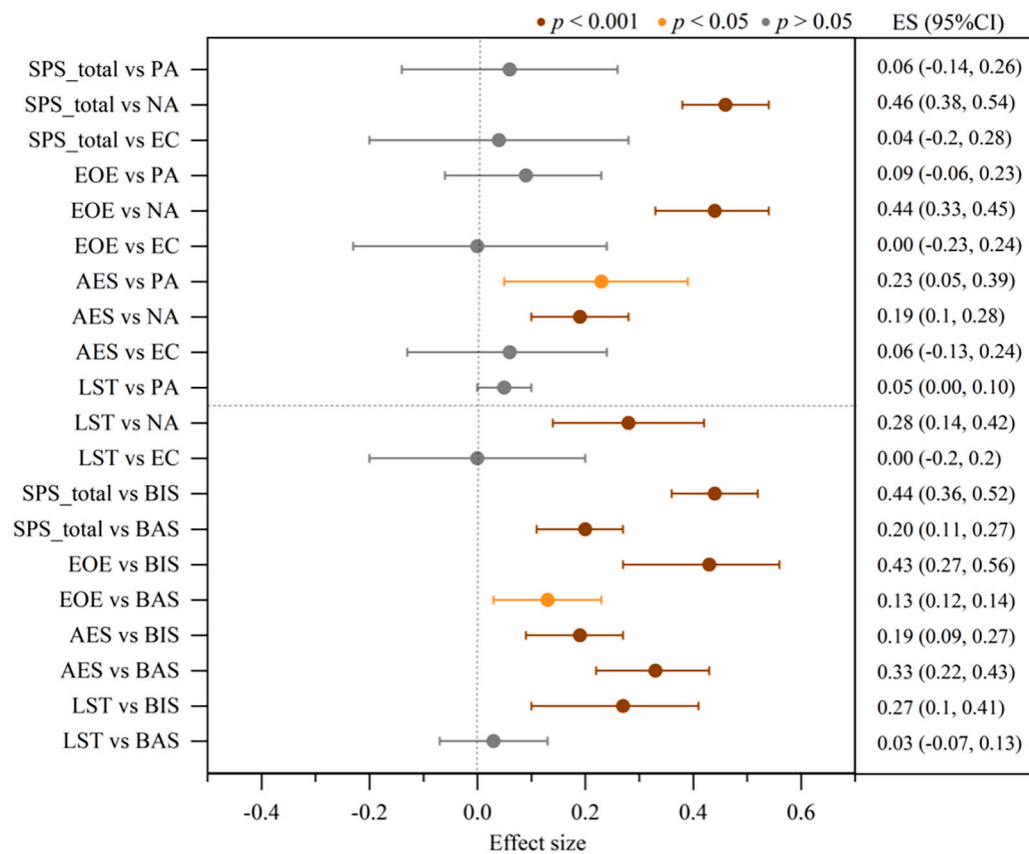


Fig. 4. The forest plot illustrating the correlation between the SPS model and temperament traits.

= 0.28), followed by Asians ($r = 0.11$), and showed no significant correlation in North Americans ($r = -0.17$, $Q_b = 22.07$, $p < 0.05$). The AES-BAS correlation was also stronger in Europeans ($r = 0.43$) than in Asians ($r = 0.32$), with no significant correlation in North American ($r = 0.11$, $Q_b = 7.30$, $p < 0.05$).

No age or regional differences were found for the correlations between the SPS total score/EOE/AES/LST and BIS.

3.4. Network analysis: Community detection in sensitivity trait network

Results from the community detection revealed three optimal community modules (Fig. 5). The median network structure had three modules (CI 95 % [3.10, 4.90]). With the frequencies, three modules were found 69 % of the time or in 345 of 500 bootstrap replicates, while four modules were found 31 % of the time or in 155 of 500 bootstrap replicates (Christensen et al., 2020). Specifically, Module 1 comprised the SPS subdimensions and openness. Module 2 included negative affect, effortful control, neuroticism and BAS. Module 3 encompassed traits such as positive affect, extraversion and BIS. The network loadings of all nodes are presented in Table S5 (Appendix B). The sensitivity analysis demonstrated that the network structure was highly robust. The estimated network models, including the number and specific placement of edges, were identical across all tested values of the EBICglasso tuning parameter ($\gamma = 0, 0.5, 1.0$, and 2.0). This stability indicates that the meta-analytic correlation matrix provided a highly robust signal, and confirms that our findings are not dependent on the specific penalty value chosen.

3.4.1. Age-specific networks

The analysis of age-specific networks revealed that the optimal number of modules was determined to be two for both the children/adolescents (CI 95 % [1.04, 2.96]) and adults (CI 95 % [1.91, 2.09])

network, with the modules found in 68 % of bootstrap replicates for children/adolescents and 99 % for adults.

In the children/adolescents network (see Fig. 6a), Module 1 consisted of four nodes: EOE, LST, neuroticism, and BIS. Module 2 also contained four nodes: AES, BAS, extraversion, openness (Fig. 6a). In the adult network (Fig. 6b), Module 1 comprised five nodes: EOE, LST, BIS, neuroticism, and extraversion, while Module 2 included three nodes: AES, openness, and BAS. The network loadings for each node across both age networks are presented in Table S6 (Appendix B).

3.4.2. Region-specific networks

Similar to the age networks, the optimal number of modules in the three region-based networks was found to be two, with 99 % of bootstrap replicates for Europe, 100 % for Asia, and 89 % for North America. The corresponding modular stability coefficients with 95 % confidence intervals were [1.77, 2.23] for Europe, [1.99, 2.01] for Asia, and [1.40, 2.60] for North America.

In the European network (see Fig. 7a), module 1 consisted of five nodes: EOE, LST, neuroticism, extraversion, BIS. Module 2 primarily included AES, openness, and BAS. In the Asian network (see Fig. 7b), Module 1 similarly included EOE, LST, neuroticism, and BIS, but unlike the European network, extraversion was not part of Module 1. Instead, extraversion formed Module 2, along with AES, openness, and BAS. In the North American network (see Fig. 7c), Module 1 clustered the three sub-dimensions of SPS (EOE, LST, AES) with BIS and neuroticism. Module 2 grouped BAS, extraversion, and openness. The network loadings of the three regional networks can be found in Table S7 (Appendix B).

4. Discussion

SPS is widely recognized as a marker for environmental sensitivity,

Table 2

The significant moderating test results of age and region.

Paired Variables	Moderator	Heterogeneity Test		Categories	k	N	Effect size(r)	95 % CI	
		Q _b	df p					LL	UL
SPS (its subdimensions)									
EOE vs LST	Age	18.07***	1	Adult	22	10,013	0.63***	0.57	0.68
				Child	16	11,500	0.45***	0.39	0.50
SPS and Personality									
SPS vs N	Age	11.37***	1	Adult	15	7199	0.52***	0.46	0.57
				Child	9	5989	0.35***	0.26	0.43
SPS vs O	Age	5.12*	1	Adult	15	9716	0.17***	0.10	0.23
				Child	9	5989	0.05	−0.01	0.12
EOE vs N	Age	13.57***	1	Adult	15	9716	0.65***	0.57	0.72
				Child	9	5989	0.37***	0.22	0.50
EOE vs E	Region	16.11**	2	Asia	5	5114	−0.03	−0.52	0.47
				Europe	18	8824	−0.29***	−0.35	−0.22
				North America	1	851	−0.09**	−0.16	−0.02
				Asia	4	3649	0.20*	0.03	0.35
AES vs E	Region	9.22**	2	Europe	18	9557	0.13***	0.06	0.20
				North America	1	851	0.00	−0.07	0.07
				Child	9	5989	0.22***	0.17	0.26
SPS and Temperament									
SPS vs NA	Region	18.62***	2	Asia	2	1408	0.30***	0.25	0.35
				Europe	10	3328	0.50***	0.35	0.62
				North America	7	1681	0.44***	0.39	0.49
EOE vs NA	Region	4.08*	1	Asia	2	1408	0.31***	0.27	0.36
				Europe	6	1724	0.49***	0.33	0.63
				North America	0	—	—	—	—
EOE vs PA	Region	17.39***	1	Asia	2	1408	−0.04	−0.09	0.01
				Europe	2	592	0.22***	0.11	0.33
				North America	0	—	—	—	—
LST vs NA	Region	5.88*	1	Asia	2	1408	0.09***	0.03	0.14
				Europe	6	1724	0.36***	0.15	0.54
				North America	0	—	—	—	—
AES vs NA	Region	7.36**	1	Asia	2	1408	0.06	−0.01	0.13
				Europe	8	2683	0.23***	0.13	0.32
				North America	0	—	—	—	—
SPS vs BAS	Age	4.60*	1	Adult	5	2384	0.11*	0.02	0.20
				Child	5	2334	0.28***	0.15	0.40
EOE vs BAS	Region	22.07*	2	Asia	4	2128	0.11**	0.03	0.19
				Europe	2	592	0.28***	0.20	0.35
				North America	1	118	−0.17	−0.34	0.01
LST vs BAS	Age	4.23*	1	Adult	2	838	0.12***	0.06	0.19
				Child	4	2000	−0.02	−0.13	0.10
AES vs BAS	Region	7.30*	2	Asia	4	2469	0.32***	0.19	0.44
				Europe	2	592	0.43***	0.27	0.57
				North America	1	118	0.11	−0.21	0.02

Note. *Q_b*, Cochran's measure of heterogeneity of subgroups; *k*, number of pooled effect size (*k* = 0 indicates no available data for this category); *N*, total sample size; *r*, correlation coefficient; 95 % CI, 95 % confidence intervals for corrected correlation coefficients; SPS, Sensory processing sensitivity; EOE, ease of excited; LST, low sensory threshold; AES, aesthetic sensitivity; BAS, behavioral approach system; PA, positive affectivity; EC, effortful control; O, openness; E, extroversion; BIS, behavioral inhibition system; NA, negative affectivity; N, neuroticism.

* *p* < 0.05.

** *p* < 0.01.

*** *p* < 0.001.

yet questions remain about its uniqueness and overlap with established temperament and personality traits (Aron & Aron, 1997; Pluess, 2015). To address this, our study conducted the most comprehensive meta-analysis to date, creating a robust empirical map of SPS's associations with key traits. Acknowledging that this evidence is derived primarily from the literature's use of the HSPS, our network analysis nonetheless offered compelling insights into the nature of the SPS construct itself. This study yielded three key findings. First, SPS emerged as a relatively distinct environmental sensitivity trait, as revealed through the network community detection. This structural independence stems from SPS embodying a unique psychometric profile that combines sensitivities to both negative stimuli and positive aesthetic experiences. This specific dual-valence pattern not only distinguishes it from univalent constructs like Neuroticism but also validates its role as a marker for broader Environmental Sensitivity (Greven et al., 2019). Second, in demonstrating this unique profile, the study validates key characteristics of

SPS, thought to be reflecting heightened emotional reactivity and deeper cognitive processing. Third, the relations and clustering of traits varied by age and region; for instance, traits such as AES and extraversion shifted between network modules depending on these factors. Ultimately, by clarifying its structural position, this work provides a more stable and nuanced foundation for future research into the developmental and neural underpinnings of SPS.

4.1. SPS as an environmental sensitivity trait: A unique trait with both "dark" and "bright" sides

This study expands on Lionetti et al. (2019) by conducting the most comprehensive meta-analysis to date, examining SPS's structural associations with sensitivity-related traits. Network analysis confirmed SPS as an independent trait, forming a distinct module within the broader trait network. This supports its theoretical validity amid ongoing

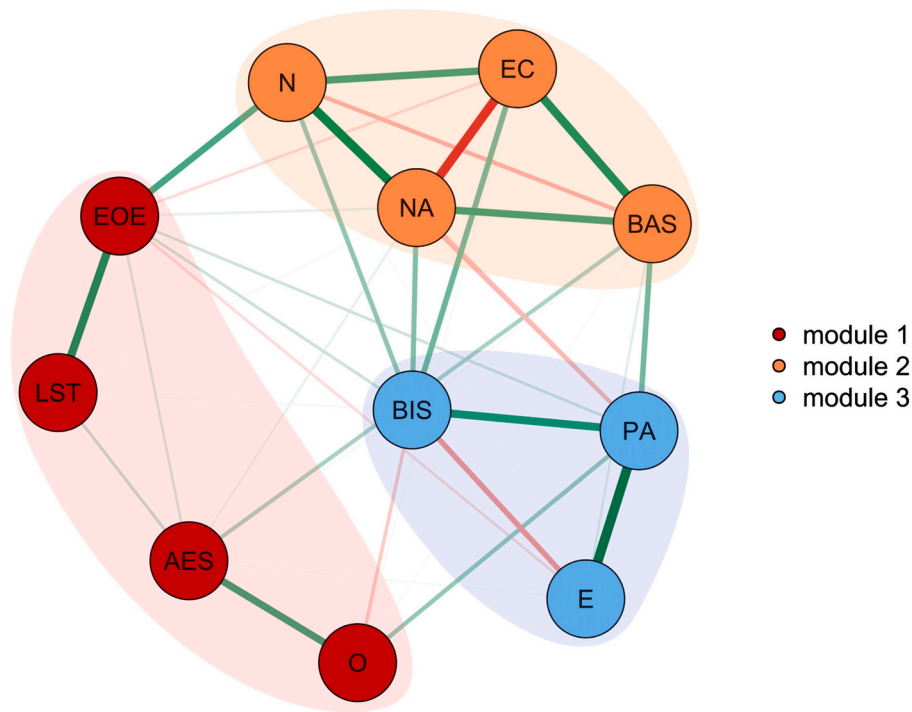


Fig. 5. Community detection of the comprehensive sensitivity traits network

Note. Green lines represent positive correlations between nodes, while red lines represent negative correlations. Thicker lines indicate stronger correlations, and thinner lines indicate weaker correlations. EOE, ease of excited; LST, low sensory threshold; AES, aesthetic sensitivity; PA, positive affectivity; NA, negative affectivity; EC, effortful control; O, openness; E, extroversion; N, neuroticism; BIS, behavioral inhibition system; BAS, behavioral approach system.

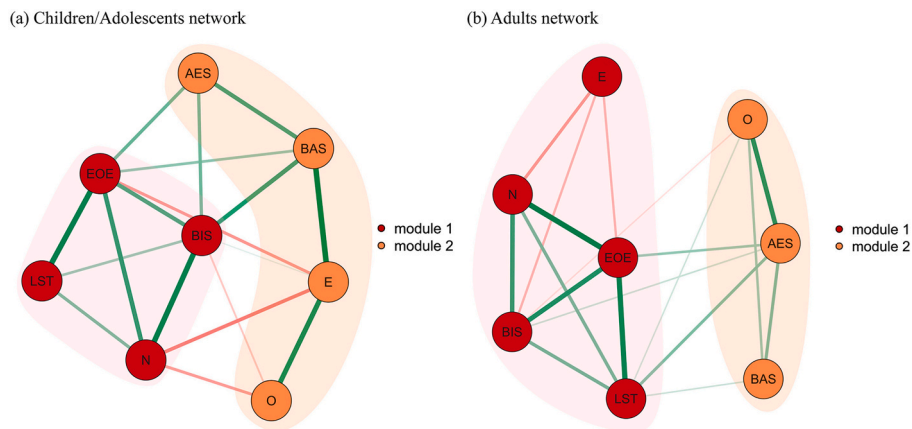


Fig. 6. Community detection of the children/adolescents and adults sensitivity traits networks

(a) Children/Adolescents network (b) Adults network

Note. Green lines represent positive correlations between nodes, while red lines represent negative correlations. Thicker lines indicate stronger correlations, and thinner lines indicate weaker correlations. EOE, ease of excited; LST, low sensory threshold; AES, aesthetic sensitivity; O, openness; E, extroversion; N, neuroticism; BIS, behavioral inhibition system; BAS, behavioral approach system.

debates about its overlap with established personality traits (Bröhl et al., 2022; Hellwig & Roth, 2021). Furthermore, although there have been studies attempting to explore the neural underpinnings of SPS, such as findings that SPS is associated with increased activation in brain regions involved in higher-order visual processing, as well as the insula and prefrontal cortex, even after controlling for traits such as neuroticism and introversion (Acevedo et al., 2014; Jagiellowicz et al., 2011). By clarifying SPS's independence, this study helps solidify its role in understanding individual-environment interactions and encourages further research into its neural mechanisms, which have been limited due to uncertainty about its independence.

Both the meta-analytic correlations and network connections reveal

SPS's dual nature, encompassing both "dark" and "bright" sides. Consistent positive and negative sensitivity modules observed across age groups and regions corresponds to the vulnerability and plasticity outlined in the differential susceptibility theory. On the "dark" side, EOE and LST correlate strongly with neuroticism and BIS, highlighting heightened vulnerability to stress and anxiety in adverse environments (Andeweg et al., 2021; Benham, 2006; Slagt et al., 2018). Conversely, the "bright" side, represented by AES, correlates with openness and BAS, suggesting enhanced benefits from supportive environments (e.g., warm parenting, high peer support and acceptance, Jagiellowicz et al., 2016). Collectively, the dual nature of SPS suggests that whether the "dark" side or the "bright" side is amplified depends heavily on environmental

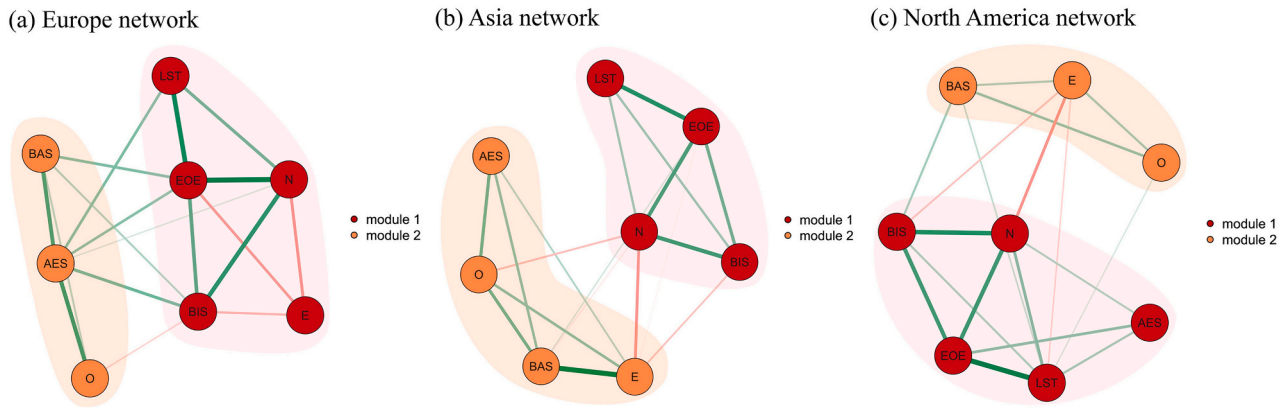


Fig. 7. Community detection of the regional sensitivity traits networks

(a) Europe network (b) Asia network (c) North America network

Note. Green lines represent positive correlations between nodes, while red lines represent negative correlations. Thicker lines indicate stronger correlations, and thinner lines indicate weaker correlations. EOE, ease of excited; LST, low sensory threshold; AES, aesthetic sensitivity; O, openness; E, extroversion; N, neuroticism; BIS, behavioral inhibition system; BAS, behavioral approach system.

factors (Ellis & Boyce, 2008). Future research should further explore how specific environmental factors moderate the expression of SPS's dual nature, as well as the implications for interventions.

Furthermore, SPS shows stronger connections to its “dark” side, with EOE and LST consistently associated with negative outcomes, while AES exhibits weaker but positive correlations. This imbalance may partially reflect limitations in existing SPS measurement tools, which tend to emphasize negative dimensions of sensitivity while underrepresenting its potential for positive emotional and aesthetic experiences (Smolewska et al., 2006). Future studies should refine assessments by incorporating behavioral observations or multi-method approaches to better capture SPS's full spectrum, enabling strategies to harness its adaptive potential while mitigating vulnerabilities.

4.2. New insights on the features of highly sensitive individuals

Prior research has established SPS as a trait characterized by the interaction between depth of processing and greater emotional reactivity, leading to both heightened awareness of subtleties and susceptibility to overstimulation (Aron et al., 2012; Greven et al., 2019). This study contributes valuable insights by utilizing meta-analysis and network analysis methods to explore the complex relations between traits, thereby providing a robust framework for defining the characteristics of highly sensitive individuals.

First, emotional reactivity to both positive and negative environmental contexts is a core characteristic of individuals high in SPS. The strong association between SPS and neuroticism supports earlier findings that highly sensitive individuals experience greater emotional volatility and stress reactivity (Aron et al., 2012; Aron & Aron, 1997). Simultaneously, the positive correlations between AES, openness and positive affectivity suggest that these individuals also derive deeper pleasure from aesthetically pleasing stimuli, a response likely supported by increased neural activity related to reward, emotion, and memory processing (Acevedo et al., 2014, 2018). However, this deeper engagement with stimuli can also lead to quicker fatigue and overwhelm. Consequently, highly sensitive individuals may require additional downtime to recover and integrate these experiences, highlighting the inherent complexity of their strengths and vulnerabilities. Furthermore, the relation between emotional reactivity, both positive and negative, raises important questions about emotional regulation. While high-SPS individuals may face challenges in balancing emotional highs and lows, their ability to manage or modulate these intense responses over time is less understood. Given that individuals with higher sensitivity need more strategies for managing emotional intensity to prevent

overwhelm (Aron et al., 2012), future research should investigate how high-SPS individuals regulate their greater emotional responses over time and in various contexts.

Second, behavioral inhibition tendency is another key characteristic, with SPS showing a strong relation with BIS across all dimensions, indicating that high-SPS individuals are more inclined toward “inhibitory” behaviors rather than impulsive or “outbursting” reactions. BIS reflects a tendency toward cautiousness and hesitation, driven by deeper cognitive processing (McNaughton & Gray, 2000). This aligns with the core characteristics of SPS, particularly the depth of cognitive processing, where individuals often “pause and check” before acting when faced with overwhelming or threatening stimuli (Aron & Aron, 1997), potentially resulting in withdrawal or inhibition behaviors. Neurocognitive studies suggest that the BIS-related personality trait involves heightened threat evaluation, environmental scanning, and memory recall (Kennis et al., 2013). Although SPS shows a weak correlation with BAS, this dual association may reflect the ambivalent tendencies of highly sensitive individuals. Future research could explore this ambivalence more deeply or investigate whether there are subgroups within highly sensitive individuals.

4.3. A critical perspective on the SPS construct and its measurement

Our findings, by providing clear estimates of shared variance, also enable a more critical evaluation of the SPS construct and its measurement. For example, the strong, meta-analytic correlation we observed between the HSPS-EOE subscale and Neuroticism ($r = 0.58$) is particularly noteworthy. This considerable overlap (34 % shared variance), combined with the face validity of items like “Do you startle easily?”, gives credence to criticisms that the EOE subscale is largely redundant with Neuroticism. However, a 34 % overlap also implies that 66 % of the variance in EOE is not shared with Neuroticism. This raises a critical question: What does this remaining, unshared variance represent? SPS theory suggests it reflects a broader, underlying sensitivity that also drives AES and LST. This interpretation is supported by our network analysis, where the three SPS facets consistently formed their own distinct community, indicating they are bound by shared information beyond what they owe to Neuroticism or Openness. Therefore, our results reframe the debate. The question is not simply whether SPS is “unique”, but rather how to best conceptualize its structure. One possibility, consistent with theory, is that SPS represents a relatively distinct temperament trait; another is that it reflects a recognizable profile that substantially overlaps with established traits while also encompassing additional variance, with some scholars further suggesting that it may

align with a specific profile of lower-level personality traits at the Big Five facet level (Bröhl et al., 2022). By robustly quantifying these relationships, our meta-analysis provides the quantitative foundation upon which these theoretical and measurement-related questions can now be more rigorously debated.

4.4. Age groups differences in high SPS individuals

Across age groups, changing relations between SPS and personality and temperament traits across the lifespan may shape the gradual evolution of sensitivity expression in individuals with high SPS. In childhood and adolescence, SPS shows a stronger association with BAS than in adulthood, possibly due to developmental immaturity, such as the incomplete development of the dopamine system, the prefrontal cortex, and lower gray matter volumes in the ventral striatum (Adrián-Ventura et al., 2019). High-SPS youth may be more inclined to approach positive stimuli or immediate rewards, as external exploration often provides positive internal experiences. Additionally, during this developmental stage, sensitivity trait networks are divided into two modules: one comprising traits associated with negative outcomes (e.g., EOE, LST, neuroticism, BIS), and another including traits linked to positive outcomes (e.g., AES, openness, BAS, extraversion). While SPS correlates with neuroticism, this association is weaker in childhood and adolescence than in adulthood, suggesting emotional instability in high-SPS individuals begins to emerge but is less pronounced, setting a baseline for persistent vulnerability to negative emotions in adulthood.

In adulthood, the correlation between SPS and neuroticism strengthens, reflecting more persistent emotional instability. However, a positive relation between SPS and openness also emerges, allowing highly sensitive adults to process environmental cues more deeply and enrich their emotional experiences (Bridges & Schendan, 2019; Lionetti et al., 2019). Despite SPS's vulnerabilities, this heightened sensitivity also fosters personal growth and aesthetic appreciation. Notably, extraversion shifts from a relatively positive module (e.g., AES, BAS, openness) in the children/adolescents network to a relatively negative sensitivity module (e.g., EOE, neuroticism, BIS) in the adult network. For highly sensitive adults, extraversion can become a source of emotional strain, as overstimulation or social fatigue diminishes its positive effects on well-being (Specht et al., 2014; Wilt & Revelle, 2009). This suggests that extraversion's impact may depend on the individual's capacity to manage social and environmental demands, which may be particularly challenging for high-SPS adults.

4.5. Regional differences: Distinct features in Europe, America, and Asia

The relations between SPS and personality or temperament vary significantly across regions. Broadly, individuals in Europe display heightened emotional reactivity to both positive and negative stimuli. In contrast, those in Asia emphasize negative emotion regulation, while North America exhibit a unique clustering of AES with traits related to negative sensitivity. These distinctions provide insights into how SPS is expressed in diverse cultural contexts.

4.5.1. Europe

European individuals high in SPS show stronger emotional responses to both positive and negative stimuli compared to their Asian and North American counterparts, consistent with individualistic cultural norms that prioritize self-reflection and emotional expression (Chentsova-Dutton et al., 2010). While this emphasis can foster emotional depth, it also increases the intensity of emotional reactivity (De Leersnyder et al., 2013; Grossmann & Kross, 2010). This is particularly true for high-SPS individuals, who, due to their heightened emotional reactivity and increased cognitive processing of emotional experiences, are more prone to stronger emotional reactions. Furthermore, Europe and North America are considered as "Western cultures" when discussing differences with highly sensitive individuals from other cultures (Aron et al.,

2010; Kurczewska et al., 2024). However, unlike North America, Europe exhibits a stronger association between SPS and negative affectivity, albeit modest in magnitude. This finding challenges the traditional "West vs. East" dichotomy, highlighting nuanced variations within Western regions.

Additionally, European high-SPS individuals show strongest correlations between SPS and BAS, suggesting potential proactive pursuit of positive experiences to counterbalance negative reactivity (Carver & White, 1994; Elliot & Thrash, 2002). However, while BAS-driven behaviors may initially help regulate negative emotions, the alignment of extraversion with negative sensitivity traits in the European network suggests a potential downside: this BAS-driven behaviors may also result in overstimulation or social fatigue, further amplifying the emotional volatility that high-SPS individuals already experience. Therefore, while BAS-related behaviors offer a temporary respite, they may not always provide long-term emotional stability for SPS individuals, particularly when overstimulation becomes a risk factor (Aron et al., 2012).

4.5.2. Asia

Compared to Europe, SPS, EOE, and LST exhibit weaker positive correlations with negative affectivity in Asia, while AES remains uncorrelated. This pattern may reflect cultural norms in many Asian societies emphasizing emotional restraint and social harmony (Hofstede, 2001), Asians high in SPS may suppress negative emotions in social contexts, preventing spillover into positive emotional domains like AES, thus maintaining clearer boundaries between the negative and positive emotional experiences, whereas in European cultures, there might be more overlap between both positive and negative emotional states.

Interestingly, Compared to Europe and North America, AES in the Asian group shows a significantly stronger positive correlation with extraversion and clusters together within the same positive sensitivity module, while EOE shows no significant negative correlation with extraversion in the Asian group. These findings indicate that highly sensitive individuals in Asia are more likely to exhibit approach-oriented behaviors, whereas those in Europe and North America tend to show lower extraversion and withdrawal tendencies. Specifically, in Asian cultures, where collectivism, modesty, and social cohesion are prioritized (Nisbett & Masuda, 2003), high-SPS individuals may be more inclined to seek and engage in positive social interactions without experiencing the overstimulation often associated with extraversion in Europe and North America. Specifically, extraversion in these cultures can manifest through socially adaptive behaviors, such as maintaining harmonious relations or fulfilling social roles. This allows high-SPS individuals to better manage emotional overload while still participating in collective social environments.

4.5.3. North America

SPS is strongly positively correlated with negative affectivity in North America, second only to Europe. However, research on the correlations between negative affectivity and the three SPS subdimensions (EOE, LST, and AES) is lacking in North America, limiting comparisons across dimensions. Notably, unlike in Europe and Asia, EOE in North America does not correlate with BAS, and AES is unassociated with both BAS and extraversion. This suggests that high-SPS individuals in North America may not tend to employ approach-oriented strategies, such as seeking external experiences, as for coping with their greater emotional reactivity. Additionally, although weaker than in Europe, a negative correlation exists between EOE and extraversion, indicating that those with high EOE may experience social engagement and external stimulation as overwhelming, leading to withdrawal tendencies (Aron & Aron, 1997).

Interestingly, unlike in Europe and Asia, where AES clusters with positive-oriented traits like openness, in North America, AES clusters with EOE and LST. While AES is typically viewed as a positive facet associated with sensitivity to positive stimuli, offering emotional relief (Aron & Aron, 1997), its deeper cognitive processing component (Pluess

et al., 2018; Smolewska et al., 2006) may also amplify engagement with negative stimuli, heightening negative reactivity (Acevedo, 2020). This North American clustering pattern suggests that AES may be more closely related to sensory overload and negative reactivity than to positive emotional experiences. Overall, SPS in North America appears to be less clearly associated with positive traits, contrasting with Europe and Asia where positive and negative sensitivity coexist. These findings challenge prior classifications (Lionetti et al., 2018) and underscores the need for further research to clarify AES's role across cultures. While these patterns reflect group averages, individual variability remains substantial, highlighting the importance of context in understanding SPS expression.

4.6. Limitations and future directions

The findings of this network-based meta-analysis should be interpreted with caution due to several limitations. First, a primary limitation of this meta-analysis—and indeed, of the SPS literature as a whole—is its heavy reliance on the HSPS as a single operationalization of the construct. Our results, while robustly summarizing the available evidence on the HSPS, are inherently constrained by the psychometric properties of this one scale. The identified dimensions of EOE, LST, and AES are derived from the HSPS item pool, and it remains an open question whether this three-factor structure is a fundamental property of the SPS construct itself or an artifact of the HSPS's specific item content. Future research must prioritize the development and validation of alternative SPS measures to establish convergent validity and determine whether the observed structure is a generalizable feature of the SPS construct itself or an artifact of the specific HSPS questionnaire. Second, the sample composition is uneven, with 60.8 % European, 21.3 % North American, and only 17.9 % Asian participants, making any comparisons exploratory. Additionally, missing data in Asian and North American groups restricted the moderation effect analysis, making it difficult to assess regional and developmental differences fully. Second, all studies included were cross-sectional, preventing identification of critical developmental transitions (e.g., AES shifts or regulatory cluster differentiation). Longitudinal studies are needed to pinpoint these turning points and guide future intervention efforts. Finally, substantial heterogeneity was observed, possibly reflecting true regional or developmental variations. However, caution must be exercised regarding the influence of highly heterogeneous data on the results. For example, while most studies found a moderate negative correlation between negative affectivity and effort control, Pluess et al. (2018) reported a strong positive correlation, which diminished the overall effect size. Nevertheless, we had no grounds to exclude these data after conducting a quality evaluation. The conflicting findings contributed to discrepancies between the combined effect sizes and those of individual studies. Such differences may stem from variations in sample selection or measurement, so caution is needed when generalizing these results.

5. Conclusion

Based on 107 independent samples, this study employed multiple methods to validate the independence and structural attributes of SPS. The findings suggest that SPS is a unique environmental sensitivity marker, characterized by dual susceptibility to both positive and negative environment. Furthermore, given the increasing complexity of emotional and behavioral regulation mechanisms in high-SPS individuals from childhood and adolescence to adulthood, as well as the unique characteristics observed across different cultural contexts, longitudinal and cross-cultural studies are needed to further investigate and explore these findings. Particularly notable is that AES, as the representative of the plasticity and positive side of SPS, is generally inadequately captured by using the current tools. Future research should prioritize the refinement of assessment tools and a deeper exploration of AES to better understand its implications within the broader framework

of SPS.

CRediT authorship contribution statement

Renjiao Tang: Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Xin Dai:** Validation, Resources, Investigation, Data curation, Conceptualization. **Corina U. Greven:** Writing – review & editing, Supervision, Resources. **Zhi Li:** Conceptualization, Resources, Writing – original draft, Writing – review & editing. **Chunhong Zhu:** Investigation, Data curation. **Ni Yan:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Ethical considerations

This study is a meta-analytic review involved a secondary analysis of existing and anonymized data, no ethical approval was needed from an Institutional Review Board (IRB).

Funding statement

This research was supported by the National Social Science Fund of China [Grant No. CBA200239] (PI: Ni Yan).

Declaration of competing interest

The authors declared no potential conflicts of interest respect to the research, authorship, financially or otherwise.

Acknowledgements

The authors acknowledge the support of grants from the Fundamental Research Funds for the National Social Science Fund of China (CBA200239, PI: Ni Yan). Opinions reflect those of the authors and do not necessarily reflect those of the granting agencies.

Appendix. Supplementary materials

Supplementary materials to this article can be found online at <https://doi.org/10.1016/j.paid.2025.113435>.

Data availability

This research was preregistered (September 25, 2024, with subsequent amendments) at <https://osf.io/3efmt>. All study materials, data, and analysis scripts used for this research work can be accessed online https://osf.io/kda6m/?view_only=e0836a2aea814c12a7759beb8b3104fb.

References

- Acevedo, B., Aron, E., Pospos, S., & Jessen, D. (2018). The functional highly sensitive brain: a review of the brain circuits underlying sensory processing sensitivity and seemingly related disorders. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 373(1744), 20170161. <https://doi.org/10.1098/rstb.2017.0161>
- Acevedo, B. P. (2020). *The highly sensitive brain: Research, assessment, and treatment of sensory processing sensitivity*. Elsevier Academic Press.

- Acevedo, B. P., Aron, E. N., Aron, A., Sangster, M. D., Collins, N., & Brown, L. L. (2014). The highly sensitive brain: an fMRI study of sensory processing sensitivity and response to others' emotions. *Brain and Behavior*, 4(4), 580–594. <https://doi.org/10.1002/brb3.242>
- Adrián-Ventura, J., Costumero, V., Parcet, M. A., & Ávila, C. (2019). Reward network connectivity “at rest” is associated with reward sensitivity in healthy adults: A resting-state fMRI study. *Cognitive, Affective, & Behavioral Neuroscience*, 19, 726–736. <https://doi.org/10.3758/s13415-019-00688-1>
- Ahadi, B., & Basharpour, S. (2010). Relationship between sensory processing sensitivity, personality dimensions and mental health. *Journal of Applied Sciences*, 10(7), 570–574. <https://doi.org/10.3923/jas.2010.570.574>
- Allport, G. W. (1961). *Pattern and growth in personality*. Holt, Reinhart & Winston.
- Andeweg, S. M., Bodrij, F. F., Prevoo, M. J. L., Rippe, R. C. A., & Alink, L. R. A. (2021). Does sensory-processing sensitivity moderate the effect of household chaos on caregiver sensitivity? An experimental design. *Journal of Family Psychology*, 35(3), 356–365. <https://doi.org/10.1037/fam0000766>
- Andresen, M., Goldmann, P., & Volodina, A. (2018). Do overwhelmed expatriates intend to leave? The effects of sensory processing sensitivity, stress, and social capital on expatriates' turnover intention. *European Management Review*, 15(3), 315–328. <https://doi.org/10.1111/emre.12120>
- Aron, A., Ketay, S., Hedden, T., Aron, E. N., Rose Markus, H., & Gabrieli, J. D. (2010). Temperament trait of sensory processing sensitivity moderates cultural differences in neural response. *Social Cognitive and Affective Neuroscience*, 5(2–3), 219–226. <https://doi.org/10.1093/scan/nsq028>
- Aron, E. N., & Aron, A. (1997). Sensory-processing sensitivity and its relation to introversion and emotionality. *Journal of Personality and Social Psychology*, 73(2), 345–368. <https://doi.org/10.1037/0022-3514.73.2.345>
- Aron, E. N., Aron, A., & Davies, K. M. (2005). Adult shyness: The interaction of temperamental sensitivity and an adverse childhood environment. *Personality and Social Psychology Bulletin*, 31(2), 181–197. <https://doi.org/10.1177/0146167204271419>
- Aron, E. N., Aron, A., & Jagiellowicz, J. (2012). Sensory processing sensitivity: A review in the light of the evolution of biological responsivity. *Personality and Social Psychology Review*, 16(3), 262–282. <https://doi.org/10.1177/1088868311434213>
- Belsky, J., & Pluess, M. (2009). Beyond diathesis stress: Differential susceptibility to environmental influences. *Psychological Bulletin*, 135(6), 885–908. <https://doi.org/10.1037/a0017376>
- Benham, G. (2006). The highly sensitive person: Stress and physical symptom reports. *Personality and Individual Differences*, 40(7), 1433–1440. <https://doi.org/10.1016/j.paid.2005.11.021>
- Borenstein, M., Cooper, H., Hedges, L., & Valentine, J. (2009). Effect sizes for continuous data. In H. Cooper, L. V. Hedges, & J. C. Valentine (Eds.), *The handbook of research synthesis and meta-analysis* (2nd ed., pp. 221–235). Russell Sage Foundation.
- Bridges, D., & Schendan, H. E. (2019). Sensitive individuals are more creative. *Personality and Individual Differences*, 143, 179–185. <https://doi.org/10.1016/j.paid.2018.09.016>
- Bröhl, A. S., Van Leeuwen, K., Pluess, M., De Fruyt, F., Bastin, M., Weyn, S., ... Bijttebier, P. (2022). First look at the Five-Factor Model personality facet associations with sensory processing sensitivity. *Current Psychology*, 41(8), 5034–5047. <https://doi.org/10.1007/s12144-020-00998-5>
- Butts, C. T. (2008). Social Network Analysis with sna. *Journal of Statistical Software*, 24, 1–51. <https://doi.org/10.18637/jss.v024.i06>
- Carver, C. S., & White, T. L. (1994). Behavioral inhibition, behavioral activation, and affective responses to impending reward and punishment: The BIS/BAS scales. *Journal of Personality and Social Psychology*, 67(2), 319–333. <https://doi.org/10.1037/0022-3514.67.2.319>
- Chentsova-Dutton, Y. E., Tsai, J. L., & Gotlib, I. H. (2010). Further evidence for the cultural norm hypothesis: Positive emotion in depressed and control European American and Asian American women. *Cultural Diversity & Ethnic Minority Psychology*, 16(2), 284–295. <https://doi.org/10.1037/a0017562>
- Chiao, J. Y., & Blizinsky, K. D. (2010). Culture-gene coevolution of individualism-collectivism and the serotonin transporter gene. *Proceedings of the Royal Society B: Biological Sciences*, 277(1681), 529–537. <https://doi.org/10.1098/rspb.2009.1650>
- Christensen, A. P., Golino, H., & Silvia, P. J. (2020). A Psychometric Network Perspective on the Validity and Validation of Personality Trait Questionnaires. *European Journal of Personality*, 34(6), 1095–1108. <https://doi.org/10.1002/per.2265>
- De Gucht, V., Wilderjans, T. F., Garcia, F. K., & Maes, S. (2023). Dimensionality and validation of the Highly Sensitive Person Scale (HSPS) in a Dutch general population sample and two clinical samples. *Journal of Individual Differences*, 44(3), 205–213. <https://doi.org/10.1027/1614-0001/a000394>
- De Leersnyder, J., Boiger, M., & Mesquita, B. (2013). Cultural regulation of emotion: Individual, relational, and structural sources. *Frontiers in Psychology*, 4, Article 55. <https://doi.org/10.3389/fpsyg.2013.00055>
- Elliot, A. J., & Thrash, T. M. (2002). Approach-avoidance motivation in personality: Approach and avoidance temperaments and goals. *Journal of Personality and Social Psychology*, 82(5), 804–818. <https://doi.org/10.1037/0022-3514.82.5.804>
- Ellis, B. J., & Boyce, W. T. (2008). Biological Sensitivity to Context. *Current Directions in Psychological Science*, 17(3), 183–187. <https://doi.org/10.1111/j.1467-8721.2008.00571.x>
- Epskamp, S., Borsboom, D., & Fried, E. I. (2018). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, 50(1), 195–212. <https://doi.org/10.3758/s13428-017-0862-1>
- Epskamp, S., & Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological Methods*, 23(4), 617–634. <https://doi.org/10.1037/met0000167>
- Gerstenberg, F. X. (2012). Sensory-processing sensitivity predicts performance on a visual search task followed by an increase in perceived stress. *Personality and Individual Differences*, 53(4), 496–500. <https://doi.org/10.1016/j.paid.2012.04.019>
- Golino, H., & Christensen, A. P. (2024). EGAnet: Exploratory Graph Analysis – A framework for estimating the number of dimensions in multivariate data using network psychometrics.
- Gray, J. A. (1982). *The Neuropsychology of Anxiety: an Enquiry Into The functions of the Septo-hippocampal System*. New York: Oxford University Press.
- Greven, C. U., Lionetti, F., Booth, C., Aron, E. N., Fox, E., Schendan, H. E., ... Homberg, J. (2019). Sensory Processing Sensitivity in the context of Environmental Sensitivity: A critical review and development of research agenda. *Neuroscience and Biobehavioral Reviews*, 98, 287–305. <https://doi.org/10.1016/j.neubiorev.2019.01.009>
- Grossmann, I., & Kross, E. (2010). The Impact of Culture on Adaptive Versus Maladaptive Self-Reflection. *Psychological Science*, 21(8), 1150–1157. <https://doi.org/10.1177/0956797610376655>
- Hedges, L. V., & Vevea, J. L. (1998). Fixed-and random-effects models in meta-analysis. *Psychological Methods*, 3(4), 486–504. <https://doi.org/10.1037/1082-989X.3.4.486>
- Hellwig, S., & Roth, M. (2021). Conceptual ambiguities and measurement issues in sensory processing sensitivity. *Journal of Research in Personality*, 93, 104–130. <https://doi.org/10.1016/j.jrp.2021.104130>
- Higgins, J. P., Thompson, S. G., Deeks, J. J., & Altman, D. G. (2003). Measuring inconsistency in meta-analyses. *BMJ (Clinical Research Ed.)*, 327(7414), 557–560. <https://doi.org/10.1136/bmj.327.7414.557>
- Hofstede, G. (2001). *Culture's Consequences: Comparing Values, Behaviors, Institutions, and Organizations Across Nations*. SAGE Publications.
- Homberg, J. R., Schubert, D., Asan, E., & Aron, E. N. (2016). Sensory processing sensitivity and serotonin gene variance: Insights into mechanisms shaping environmental sensitivity. *Neuroscience & Biobehavioral Reviews*, 71, 472–483. <https://doi.org/10.1016/j.neubiorev.2016.09.029>
- Jagiellowicz, J., Aron, A., & Aron, E. N. (2016). Relationship between the temperament trait of sensory processing sensitivity and emotional reactivity. *Social Behavior and Personality: An International Journal*, 44(2), 185–199. <https://doi.org/10.2224/sbp.2016.44.2.185>
- Jagiellowicz, J., Xu, X., Aron, A., Aron, E., Cao, G., Feng, T., & Weng, X. (2011). The trait of sensory processing sensitivity and neural responses to changes in visual scenes. *Social Cognitive and Affective Neuroscience*, 6(1), 38–47. <https://doi.org/10.1093/scan/nsq001>
- Kennis, M., Rademaker, A. R., & Geuze, E. (2013). Neural correlates of personality: an integrative review. *Neuroscience and Biobehavioral Reviews*, 37(1), 73–95. <https://doi.org/10.1016/j.neubiorev.2012.10.012>
- Kurczewska, E., Ferensztajn-Rochowiak, E., Rybakowski, J., & Rybakowski, F. (2024). Sensory Processing Sensitivity as a trait of temperament - evolutionary, socio-cultural, biological context and relation to mental disorders. *Psychiatria Polska*, 58(2), 249–264. <https://doi.org/10.12740/PP/160301>
- Lionetti, F., Aron, A., Aron, E. N., Burns, L. G., Jagiellowicz, J., & Pluess, M. (2018). Dandelions, tulips and orchids: Evidence for the existence of low-sensitive, medium-sensitive, and high-sensitive individuals. *Translational Psychiatry*, 8(24). <https://doi.org/10.1038/s41398-017-0090-6>
- Lionetti, F., Pastore, M., Moscardino, U., Nocentini, A., Pluess, K., & Pluess, M. (2019). Sensory Processing Sensitivity and its association with personality traits and affect: A meta-analysis. *Journal of Research in Personality*, 81, 138–152. <https://doi.org/10.1016/j.jrp.2019.05.013>
- Matsumoto, D., Yoo, S. H., & Fontaine, J. (2008). Mapping expressive differences around the world: The relationship between emotional display rules and individualism versus collectivism. *Journal of Cross-Cultural Psychology*, 39(1), 55–74. <https://doi.org/10.1177/0022022107311854>
- McNaughton, N., & Gray, J. A. (2000). Anxiolytic action on the behavioural inhibition system implies multiple types of arousal contribute to anxiety. *Journal of Affective Disorders*, 61(3), 161–176. [https://doi.org/10.1016/S0165-0327\(00\)00344-X](https://doi.org/10.1016/S0165-0327(00)00344-X)
- Nisbett, R. E., & Masuda, T. (2003). Culture and point of view. *Proceedings of the National Academy of Sciences*, 100(19), 11163–11170. <https://doi.org/10.1073/pnas.1934527100>
- Nocentini, A., Menesini, E., & Pluess, M. (2018). The personality trait of environmental sensitivity predicts children's positive response to school-based antibullying intervention. *Clinical Psychological Science*, 6(6), 848–859. <https://doi.org/10.1177/2167702618782194>
- Patton, G. C., Sawyer, S. M., Santelli, J. S., Ross, D. A., Afifi, R., Allen, N. B., ... Viner, R. M. (2016). Our future: A Lancet commission on adolescent health and wellbeing. *The Lancet*, 387(10036), 2423–2478. [https://doi.org/10.1016/S0140-6736\(16\)00579-1](https://doi.org/10.1016/S0140-6736(16)00579-1)
- Pluess, M. (2015). Individual differences in environmental sensitivity. *Child Development Perspectives*, 9(3), 138–143. <https://doi.org/10.1111/cdep.12120>
- Pluess, M., Assary, E., Lionetti, F., Lester, K. J., Krapohl, E., Aron, E. N., & Aron, A. (2018). Environmental sensitivity in children: Development of the Highly Sensitive Child Scale and identification of sensitivity groups. *Developmental Psychology*, 54(1), 51–70. <https://doi.org/10.1037/dev0000406>
- Pluess, M., & Belsky, J. (2013). Vantage sensitivity: Individual differences in response to positive experiences. *Psychological Bulletin*, 139(4), 901–916. <https://doi.org/10.1037/a0030196>
- Pluess, M., & Boniwell, I. (2015). Sensory-Processing Sensitivity predicts treatment response to a school-based depression prevention program: Evidence of Vantage Sensitivity. *Personality and Individual Differences*, 82, 40–45. <https://doi.org/10.1016/j.paid.2015.03.011>
- Redfearn, R. A., van IJssum, K. W., & Stenmark, C. K. (2020). The impact of sensory processing sensitivity on stress and burnout in nurses. *International Journal of Stress Management*, 27(4), 370–379. <https://doi.org/10.1037/str0000158>

- Rothbart, M. K., & Jones, L. B. (1998). Temperament, self-regulation, and education. *School Psychology Review*, 27(4), 479–491. <https://doi.org/10.1080/02796015.1998.12085932>
- Schmidt, F. L., & Hunter, J. E. (2014). *Methods of meta-analysis: Correcting error and bias in research findings* (3rd ed.). Thousand Oaks, CA: Sage.
- Şengül-İnal, G., Kirimer-Aydinli, F., & Sümer, N. (2018). The Role of Attachment Insecurity and Big Five Traits on Sensory Processing Sensitivity. *The Journal of Psychology*, 152(7), 497–514. <https://doi.org/10.1080/00223980.2018.1482255>
- Şengül-İnal, G., & Sümer, N. (2020). Exploring the Multidimensional Structure of Sensory Processing Sensitivity in Turkish Samples. *Current Psychology*, 39(1), 194–206. <https://doi.org/10.1007/s12144-017-9751-0>
- Slagt, M., Dubas, J. S., van Aken, M. A. G., Ellis, B. J., & Deković, M. (2018). Sensory processing sensitivity as a marker of differential susceptibility to parenting. *Developmental Psychology*, 54(3), 543–558. <https://doi.org/10.1037/dev0000431>
- Smolewska, K. A., McCabe, S. B., & Woody, E. Z. (2006). A psychometric evaluation of the Highly Sensitive Person Scale: The components of sensory-processing sensitivity and their relation to the BIS/BAS and “Big Five”. *Personality and Individual Differences*, 40(6), 1269–1279. <https://doi.org/10.1016/j.paid.2005.09.022>
- Sobocko, K., & Zelenski, J. M. (2015). Trait sensory-processing sensitivity and subjective well-being: Distinctive associations for different aspects of sensitivity. *Personality and Individual Differences*, 83, 44–49. <https://doi.org/10.1016/j.paid.2015.03.045>
- Specht, J., Bleidorn, W., Denissen, J. J. A., Hennecke, M., Hutteman, R., Kandler, C., ... Zimmermann, J. (2014). What Drives Adult Personality Development? A Comparison of Theoretical Perspectives and Empirical Evidence. *European Journal of Personality*, 28(3), 216–230. <https://doi.org/10.1002/per.1966>
- Weyn, S., Van Leeuwen, K., Pluess, M., Lionetti, F., Goossens, L., Bosmans, G., ... Bijttebier, P. (2021). Improving the measurement of environmental sensitivity in children and adolescents: The Highly Sensitive Child Scale-21 Item Version. *Assessment*, 29(4), 607–629. <https://doi.org/10.1177/1073191120983894>
- Wilt, J., & Revelle, W. (2009). Extraversion. In M. R. Leary, & R. H. Hoyle (Eds.), *Handbook of individual differences in social behavior* (pp. 27–45). The Guilford Press.
- Yano, K., Kase, T., & Oishi, K. (2020). The associations between sensory processing sensitivity and the Big Five personality traits in a Japanese sample. *Journal of Individual Differences*, 42(2), 84–90. <https://doi.org/10.1027/1614-0001/a000332>